United International University
Department of Computer Science and Engineering
CSE 4811 / DS 4213: Natural Language Processing

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Practice Problems: Class 03

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1. **The chain rule and the Markov assumption.**

   (a) Consider the following model for the probability of a sentence.

   $$p(w_1 \dots w_n) = p(w_1)\, p(w_2 \mid w_1) \cdots p(w_n \mid w_1 \dots w_{n-1}) \qquad \text{(step 1)}$$
   $$p(w_i \mid w_1 \dots w_{i-1}) = p(w_i \mid w_{i-3}\, w_{i-2}\, w_{i-1}) \qquad \text{(step 2)}$$

   For each of the two steps, say what assumptions (if any) must be made about the distribution of sentences in order for that step to be valid. (You may assume that all the necessary conditional distributions are well-defined.)

   (b) Suppose we build a language model that makes use of a second-order Markov assumption (a trigram model), that is

   $$p(w_1 \dots w_n) = \prod_{i=1}^{n} p(w_i \mid w_{i-2}\, w_{i-1}), \qquad w_{-1} = w_0 = \texttt{<s>}.$$

   So we assume that the ith word w_i is independent of $w_1 \dots w_{i-3}$, once we condition on w_{i-2} and w_{i-1}. Give some examples in English where English grammar suggests that this independence assumption is very clearly violated.

   Toronto *CSC421 Neural Networks and Deep Learning* **2019** Midterm, Q1 · Solutions

   Columbia *COMS W4705 Natural Language Processing* **2015** Week 1 questions, Q4 · Solutions

2. **Bigram probabilities.** In this question, we assume that we train a bigram model. Remember that in the bigram case, the probability of each word w given the previous word v is defined as follows:

   $$p(w \mid v) = \frac{\text{count}(v\, w)}{\sum_{w'} \text{count}(v\, w')}.$$

   Consider the vocabulary $\{\text{love}, \text{builds}, \text{trust}, \texttt{<s>}, \texttt{</s>}\}$, where `<s>` indicates the beginning of a sentence and `</s>` the end of a sentence. Suppose that we have the following training data:

   ```
   <s> love builds </s>
   <s> trust builds </s>
   <s> builds love </s>
   ```

   (a) Compute all the bigram probabilities from the above corpus.

   (b) Compute the probabilities of the sentences `<s> trust builds love </s>`, `<s> love builds love </s>` and `<s> love builds trust </s>`.

   (c) One of the sentences above, although it is correct, is given a probability of 0. We would like to avoid this.

   i. How could we change the size of n?

   ii. What is the shortest sentence we could add to the training data?

   ETH Zurich *Natural Language Processing* **2024** Practice exam, Q2(a) to (c) · Solutions

3. **Adding order to naive Bayes.** Suppose you wish to extend the naive Bayes model to time series data. Formally, a training data point consists of a binary label Y and D sequentially ordered observations of some binary phenomenon where X_1 occurs before X_2, which occurs before X_3 and so on all the way down to the final observation X_D; each feature X_d is binary.

You decide to modify the naive Bayes assumption such that now, a feature X_d is conditionally independent of all earlier features $X_1, \ldots, X_{d-2}$ given the label Y *and* the previous feature X_{d-1}; the first feature X_1 depends on the label Y alone.

(a) Write down the expression for the joint distribution $P(X, Y)$, with $X = (X_1, \ldots, X_D)$, under your new naive Bayes model. Your answer must use the modified naive Bayes assumption described above.

(b) How many parameters would you need to learn in order to make predictions using your new naive Bayes model?

CMU *10-701 Introduction to Machine Learning* **2024** Practice midterm, Q5.1 · Solutions

4. **The order of an n-gram model.** Explain briefly what the n-gram assumption is. What are some advantages and disadvantages of having bigger or smaller n?

ETH Zurich *Natural Language Processing* **2021** Practice exam, Q1(b) · Solutions

5. **Generating text from a bigram model.** We are given the following corpus:

```
<s> language is awesome </s>
<s> language is awesome </s>
<s> language and students </s>
<s> awesome students like language </s>
```

(a) Using a bigram language model without smoothing, estimated from the corpus, generate four sentences, starting from `<s>`. To generate a sentence manually, randomly choose a real number between 0 and 1 and use the cumulative probabilities of the words in the vocabulary to select a word; repeat until `</s>` is drawn. For example, if two words have the non-zero conditional probabilities $P(\text{one} \mid \texttt{<s>}) = 0.25$ and $P(\text{two} \mid \texttt{<s>}) = 0.75$, a random number between 0 and 0.25 selects "one", and a random number between 0.25 and 1 selects "two".

(b) How does the lack of smoothing affect the novelty of the sentences generated?

(c) How does the nature of the corpus affect the lengths of the generated sentences?

Edinburgh *Foundations of Natural Language Processing* **2024** Tutorial 2, Q2 · Solutions

6. **Smoothing.** Use the corpus of question 5. Count `</s>` as a word, but not `<s>`, both in the vocabulary size V and in the unigram probabilities $P(w)$. Interpolation mixes the unigram and bigram estimates without smoothing: $P_{\text{INT}}(w \mid v) = \lambda_1 P(w) + \lambda_2 P(w \mid v)$, with $\lambda_1 + \lambda_2 = 1$.

(a) Compute $P(\text{awesome} \mid \text{is})$ and $P(\text{like} \mid \text{is})$ for (i) the bigram language model without smoothing, (ii) with add-one smoothing, and (iii) using interpolation with $\lambda_1 = \lambda_2 = 0.5$.

(b) One would normally select the values for λ_1 and λ_2 based on a held-out corpus. What are the characteristic features of a corpus that would result in $\lambda_2 \ll \lambda_1$?

(c) Mention a disadvantage of using interpolation to do smoothing, by referring to the bigrams *is awesome* and *is like*. Identify another type of smoothing that overcomes the shortcomings you described.

Edinburgh *Foundations of Natural Language Processing* **2024** Tutorial 2, Q3 · Solutions

7. **Add-k smoothing.**

(a) Compute the maximum likelihood estimate and the add-k estimate of $P(\text{zoidberg} \mid \text{doctor})$, assuming that the word doctor appears 15 times in the training corpus, zoidberg occurs 10 times, the bigram doctor zoidberg appears 5 times, the size of the training corpus is 50,000 word tokens, the size of the test corpus is 1000 word tokens, the size of the vocabulary is 20,000 word types, and $k = 0.25$. You can leave your answers as fractions.

(b) Suppose I have a text corpus and I divide it randomly to create training and development sets. I train two bigram models with add-k smoothing on the training set. In model bi-dev, I choose k to give the development set the smallest loss, and in model bi-train, I choose k to give the training set the smallest loss. In which model would I expect to get a larger k?

A. Model bi-dev will have a larger k than model bi-train.

B. Model bi-train will have a larger k than model bi-dev.
C. The two models will have the same k.
D. It could be any of the above.

Toronto *CSC401 Natural Language Computing* **2017** Final, short answer Q5

Edinburgh *Accelerated Natural Language Processing* **2024** Practice exam, Q1(c)

Words as vectors

8. **Word vectors.**
   (a) True or false: one-hot vector encodings of words must be the length of the entire vocabulary, while learned embeddings are generally much smaller.
   (b) The input embedding table E and the output layer use separate parameters. The table is used only for lookup. One backward pass looks up IDs $[2, 5, 2]$; the loss gradients with respect to the three returned vectors are $[2, 1]$, $[0, 3]$ and $[-1, 4]$, in that order. These are the only uses of the table. Write the two-component gradient of the loss for row 2 of E.

   UW *CSE 493G1 Deep Learning* **2024** Quiz 3, Q1.1 · Solutions

   Johns Hopkins *601.471 NLP: Self-supervised Models* **2026** Quiz 4, Q1(b) · Solutions

9. **Counting parameters.** Consider a feedforward language model. In this model, the probability $P(w_i \mid w_{i-n+1}, \ldots, w_{i-1})$ is given by:

$$\begin{aligned} P(w_i \mid w_{i-n+1}, \ldots, w_{i-1}) &= \text{softmax}(\mathbf{U}\mathbf{h}_2 + \mathbf{b}_2) \\ \mathbf{h}_2 &= \tanh(\mathbf{W}\mathbf{h}_1 + \mathbf{b}_1) \\ \mathbf{h}_1 &= \text{concatenate}(\mathbf{C}\mathbf{w}_{i-n+1}, \ldots, \mathbf{C}\mathbf{w}_{i-1}) \\ \mathbf{w}_i &= \text{onehot}(w_i) \quad \text{for all } i \end{aligned}$$

   In this notation, $\mathbf{U}$, $\mathbf{W}$ and $\mathbf{C}$ are matrices, while $\mathbf{b}_1$, $\mathbf{b}_2$, $\mathbf{h}_1$, $\mathbf{h}_2$ and $\mathbf{w}_{i-n+1}, \ldots, \mathbf{w}_{i-1}$ are vectors. The parameters of the model are $\mathbf{U}$, $\mathbf{W}$, $\mathbf{C}$, $\mathbf{b}_1$ and $\mathbf{b}_2$, while the remaining variables are intermediate layers computed by the network. Now consider the number of parameters required to represent this model. This number is determined by the size of the vocabulary $|V|$ (given to you by the data), the order n, and the dimension of the two hidden layers, $\mathbf{h}_1$ and $\mathbf{h}_2$, which we will denote d_1 and d_2, respectively (note that the first dimension must be divisible by $n-1$, but you can ignore this detail in your calculations).
   (a) How would you express the number of model parameters in terms of $|V|$, n, d_1 and d_2?
   (b) An effective size for the hidden dimension of a neural NLP model is often in the hundreds. Let $|V| = 20{,}000$. For n from 2 to 5, how many parameters would your model have if $d_1/(n-1) = d_2 = 100$? What if $d_1/(n-1) = d_2 = 1000$?
   (c) What do you conclude about the relative memory efficiency of classic n-gram and feedforward neural language models? If you increased n even further, what would happen?

   Edinburgh *Natural Language Understanding, Generation, and Machine Translation* **2025** Tutorial 1, Q2 · Solutions

10. **Counting and learning.**
    (a) Consider how a neural language model with a softmax output layer compares with a classic n-gram language model. Typically, we use techniques like smoothing or backoff in conjunction with n-gram models. Does the problem these techniques solve arise in the neural model? Why or why not?
    (b) Recall the neural language model architecture: the representations of the three previous words, rows of E, are joined into one vector e; the hidden units are $h = \sigma(We+b)$; and $\hat{y} = \text{softmax}(Uh)$ is the distribution for the next word. Why is it that the learned word representations for "professor" and "teach" are likely to be far apart? (It is not sufficient to say they are "semantically dissimilar"; your answer should mention how the representations are used in the prediction task.)

(c) Two prefixes differ in an earlier name but have the same final four tokens, `the key belonged to`. A deterministic fixed-window language model sees only those four tokens. With the same model parameters used for both prefixes, can adding hidden layers alone make their next-token distributions differ? Answer yes or no, explain why in one sentence, and name one input change that would expose the distinction.

Edinburgh *Natural Language Understanding, Generation, and Machine Translation* **2025** Tutorial 1, Q1(d) · Solutions

Toronto *CSC321 Introduction to Neural Networks and Machine Learning* **2017** Final, Q5 · Solutions

Johns Hopkins *601.471 NLP: Self-supervised Models* **2026** Quiz 4, Q1(c) · Solutions